

PROJECT REQUIREMENT DOCUMENT

Reference Document for Students

Project: AI Notes Generator

1. Team Members

Name	Role	Responsibility	Tools Used
S. Shehanaz	Team Leader	Full Stack + GitHub Management	VS Code, GitHub
M. Ankitha	Frontend Dev	React UI, Vite Frontend	VS Code, Vercel
P. Rekha	Backend Dev	FastAPI APIs, Gemini Integration	VS Code, Swagger UI
S. Hema Sree	Database	SQLite DB Design, Models	VS Code, DB Browser for SQLite

■ Note: Each person should have a clear responsibility. Two people should not do the same task.

2. Project Idea

What is your project?

AI Notes Generator is an AI-powered web application where users can generate notes using Artificial Intelligence with dynamic word limits. Users can save notes permanently, summarize PDFs, generate quizzes, and create multilingual notes.

What problem does it solve?

Students spend a lot of time preparing notes manually. This application helps users generate high-quality notes instantly and study faster.

Who will use it?

- Students – Generate notes and quizzes
- Professionals – Quick summaries and study materials
- Teachers – Create learning content

3. Features List

Must Have (Core Features)

- User Registration and Login
- Secure login with JWT authentication
- AI Notes Generation using Gemini API
- Dynamic word limit support
- Permanent notes storage (even after refresh/logout)
- Search notes and notes history
- Download notes as PDF/TXT
- PDF Upload and Summarization
- AI Quiz Generation from notes
- Multi-language Notes (English, Telugu, Hindi)

Nice to Have (Extra Features — if time permits)

- Dark/Light mode
- AI flashcards
- Cloud backup
- Notes sharing system

■ Note: Write Must Have features first. These MUST be completed before project submission. Nice to Have features can be added later.

4. Tech Stack

Layer	Technology	Purpose
Frontend	React.js + Vite	Build responsive UI
Backend	FastAPI (Python)	REST API development
Database	SQLite	Lightweight file-based database
Authentication	JWT	Secure login
API Documentation	Swagger UI	Auto-generated API docs
Version Control	GitHub	Code management and collaboration
AI Integration	Gemini API	Generate notes and quizzes

5. Pages / Screens

User-Facing Pages

- Home Page – Overview of features
- Register Page – User sign up form
- Login Page – Email and password login form
- Dashboard – Generate notes and manage content
- Notes History Page – View saved notes
- PDF Summary Page – Upload and summarize PDF
- Quiz Page – Generated quiz questions
- Profile Page – View user details

6. Database Tables

users

Column	Data Type	Description
user_id	INT (Primary Key)	Unique ID for each user
name	VARCHAR(100)	Full name of the user
email	VARCHAR(150)	Email address (must be unique)
password	VARCHAR(255)	Encrypted password
role	ENUM('user','admin')	Whether the account is user or admin

notes

Column	Data Type	Description
note_id	INT (Primary Key)	Unique ID for each note
user_id	INT (Foreign Key)	Reference to the user
topic	VARCHAR(150)	Notes topic
generated_note	TEXT	Generated AI content
word_limit	INT	Requested word limit
language	VARCHAR(50)	Selected language

7. API Design

Method	Endpoint	Description	Who Can Use
POST	/api/auth/register	Register a new user	Anyone

POST	/api/auth/login	Login and get token	Anyone
POST	/api/notes/generate	Generate AI notes	Logged-in User
POST	/api/pdf/upload	Upload and summarize PDF	Logged-in User
POST	/api/quiz/generate	Generate quiz	Logged-in User
GET	/api/notes/history	View notes history	Logged-in User

8. User Flows

Flow	Steps
User Registration	Open App → Click Register → Fill Details → Submit → Login
User Login	Open App → Login → Enter Credentials → Dashboard
Generate Notes	Enter Topic + Word Limit → Generate → View Notes
PDF Summary	Upload PDF → AI Summarizes → Save Notes
Quiz Generation	Open Notes → Generate Quiz → View Questions

■ Note: User flows should cover the main journeys a person takes in your app. Think from the user's perspective, not from the code perspective.

— *End of Requirement Document* —